SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – SHORT ANSWER TEST (30 Questions)

Course Code:	ITA0502	Course Title:	COMPUTER VISION
CO Assessed:	CO1 – Imitation (BL3)	Assessment:	Assessment Tool 1 – Short Answer Test
Weightage:	40% of CIA	Total Marks:	50 (10 Questions × 5 Marks)
CO4 Statement:	*Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.* (BTL 3)		
Student Name:	A. Archishman	Reg. No.:	192425341
Section / Batch:	2024	Date:	

Code of Conduct

I Archishman 19242534 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: ____________

Instructions

- This test contains 6 questions, each carrying 5 marks. Total: 30 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Overview of Computer Vision	Definition, objectives, workflow, and importance of Computer Vision	1	5
Features of Computer Vision	Feature extraction, object recognition, and image understanding	2	5
Levels of Computer Vision and Image Processing	Low-level, mid-level, and high-level image processing operations	3	5
Digital Image Fundamentals	Pixels, image representation, grayscale/color images, and image resolution	4,10	10
Image Sensing and Acquisition	Image sensors, acquisition process, and imaging devices	5	5
Sampling and Quantization	Sampling, quantization, and digital image formation	6	5
Image Formation Models	Perspective projection, illumination, and image formation process	7	5
Applications of Image Processing and Computer Vision	Healthcare, autonomous vehicles, surveillance, agriculture, robotics, and industrial automation	8,9	10
	GRAND TOTAL:		50 Marks

Annexure A – Short Answer Test

1. How does Computer Vision enable computers to interpret and understand visual information?

a) What are the main objectives of Computer Vision?

b) Mention any two real-world applications of Computer Vision.

2. What are the key features of Computer Vision that make it suitable for intelligent image analysis?

a) Define feature extraction.

b) How does object recognition improve Computer Vision systems?

3. How are the different levels of image processing used in Computer Vision?

a) What is low-level image processing?

b) Differentiate between mid-level and high-level image processing.

4. How is a digital image represented and what are its fundamental components?

a) Define a pixel.

b) How does image resolution affect image quality?

5. How does image sensing and acquisition convert a real-world scene into a digital image?

a) What is the role of an image sensor?

b) Name any two image acquisition devices.

6. How do sampling and quantization contribute to the formation of a digital image?

a) Define sampling.

b) What is the purpose of quantization in image processing?

7. How does the image formation model describe the process of capturing images?

a) What is perspective projection?

b) What is the role of illumination in image formation?

8. How are image processing and Computer Vision related, and how do they differ?

a) Define image processing.

b) State two differences between image processing and Computer Vision.

9. How is Computer Vision applied in solving real-world problems across different domains?

a) Explain one application in healthcare.

b) Explain one application in autonomous vehicles.

10. How do digital image fundamentals support Computer Vision applications?

a) Why are pixels considered the basic building blocks of digital images?

b) How do sampling and image resolution influence image analysis?

11-07-26 | Assignment Tool-I Test-1 | Slot-C Batch: 2024 | A. Archishman 192425341 ITA0502

1. Computer vision helps computers to understands Images & videos.
a. Object detection, recognition, scene understanding
b. Face Recognition, Imaging.

2. Key Features of computer vision: Recognitions, Analyzes, - -
a. Finding the important images & videos features.
b. object recognition identifies classifing the objects.

3. a. Low-level image processing :- Enhancement of the noise removal.
b. mid-Level image processing :- is Feature extraction.
High-level is object recognitions.

4. a. pixel is known as smallest unit of an image.
b. Higher resolution gives the better image quality.

5. a. Image sensor, it converts lights into the digital signals.
b. Two acquisition devices are :- Camera, scanner.

6. a. Sampling :- It is nothing but, it converts image into pixels values.
b. The quantization is assigns for intensity values.

7. a.

b. illumination provides light for image captures.

8. a. Image processing is, it enhances the better images.

b. Image processing is for improving the images.

Computer vision is for the understands the images.

9. a. Application for Healthcare is: Discease detection.

b. Autonomous vehicles for detects roads and for obstacles.

10. a. pixels are the basic unit of the digital images.

b.

Rubrics for Calculation

Criteria	Weigh tage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	50%	Demonstrates thorough, accurate understanding of concepts	Good understand ing with minor gaps	Basic understandin g with some misconceptio ns	Poor understand ing; major misconcept ions		
Presentation	50%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation, lacks depth	Poor or unclear explanation		

Faculty In-charge	Course Coordinator	Head of Department
Signature: ____________	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________